

Dataset Description and Statistics

Dataset Description

The dataset is used for a Multi-Task Leukemia Classification and Grading system. It consists of blood cell images categorized into 4 types of Leukemia, which are further mapped to 2 severity grades.

Classes (Types):

- ALL (Acute Lymphoblastic Leukemia)
- AML (Acute Myeloid Leukemia)
- CLL (Chronic Lymphocytic Leukemia)
- CML (Chronic Myeloid Leukemia)

Grades:

- Acute (Blast): Mapped from ALL and AML
- Chronic: Mapped from CLL and CML

Image Counts

The dataset is balanced with 320 images per class, totaling 1280 images.

It is split into Training (70%), Validation (15%), and Testing (15%).

- Training: 896 images (224 per class)
- Validation: 192 images (48 per class)
- Testing: 192 images (48 per class)
- Total: 1280 images (320 per class)

Size Details

Input Resolution: The system processes all images by resizing them to 224 x 224 pixels.

Channels: 3 (RGB).

Original Formats: The pipeline accepts .jpg, .png, .bmp, and .tif files before resizing them for the model (ResNet18 backbone).